Transparency in Artificial Intelligence

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Learning Outcomes

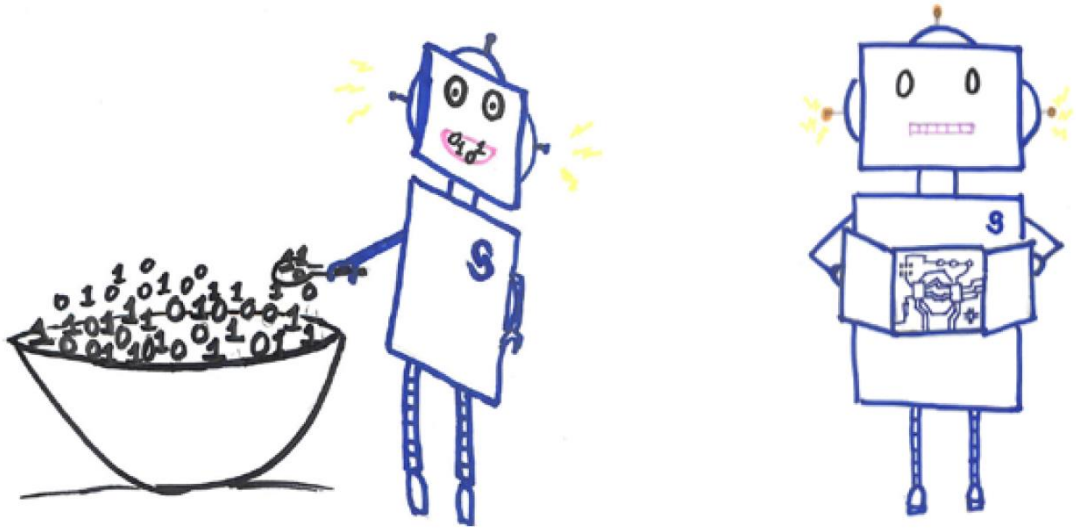
- Understand the characteristics of white box vs. black box models
- Distinguish between Interpretable and explainable AI
- Understand the workings of feature importance calculation
    - Feature permutation importance
    - SHAP
    - LIME

Typical Machine Learning Pipeline

- Clean and preprocess data
- Engineer features based on prior knowledge and experimentation
- Pick promising model families for the task
- Do hyperparameter tuning and

Voila !!! Let’s conclude the work and move on …

Job Well Done?



Accuracy Alone???

- Sometimes, accuracy alone is **not** enough !!!

- Mission critical systems need more that accuracy
  - Medicine
  - Law Enforcement
  - Finance
  - Education

Hence, Model Interpretability and Explanations

A Taxonomy

- We can categorise different model transparency techniques based on different verticals

- Allows us to map different requirements and constraints of ML tasks to align transparency methods

Black/White box models in AI

- Black box Model

A system that ***does not reveal its internal mechanisms***. In machine learning, “black box” describes models that **cannot be understood by looking at their parameters** (e.g. a neural network).

- White box Model

Opposite of black box. The internals provide a **humanly intuitive picture of how the model behaves** (e.g. Linear/ logistic Regression)

Interpretable and Explainable AI

- Interpretable AI

When the model is white box, allowing stakeholders to understand the model behaviour just by looking at the model.

- Explainable AI

What we need to do in the absence of interpretability. Having to use an additional mechanism to make the model behaviour intuitive to human stakeholders.

Interpretable and Explainable AI

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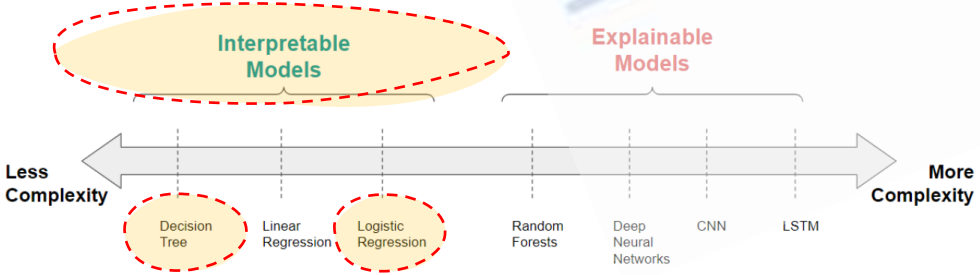
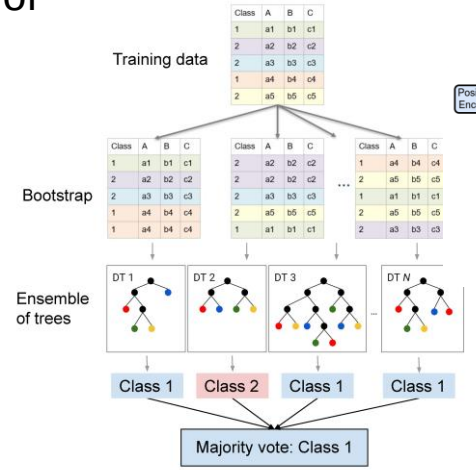
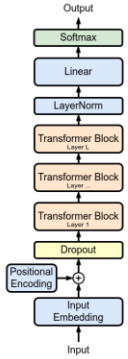
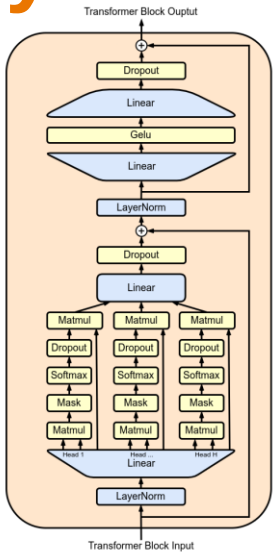


Black Box Model Zoo → Explainability

- Many models out there are black box models
- Super complicated with ensembles, billions of parameters etc…
- Explainable AI is key to allow transparency







Intrinsic vs. Post-hoc

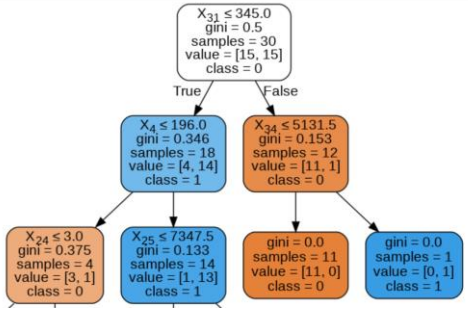
- **Intrinsic:**

Interpretability is achieved by restricting the complexity of the machine learning model

- **Post-hoc:**

Applying methods that analyse the model after training

- Aligned with White box and Black box models

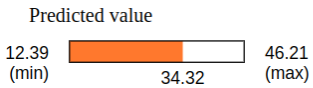


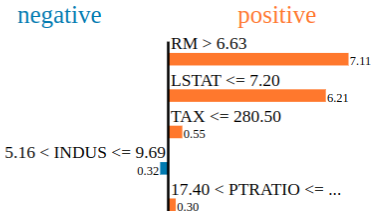

```
Call:
lm(formula = ROLL ~ UNEM + HGRAD + INC, data = datavar)

Residuals:
      Min        1Q    Median        3Q       Max
-1148.840  -489.712    -1.876   387.400  1425.753

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.153e+03  1.053e+03  -8.691 5.02e-09 ***
UNEM         4.501e+02  1.182e+02   3.809 0.000807 ***
HGRAD        4.065e-01  7.602e-02   5.347 1.52e-05 ***
INC          4.275e+00  4.947e-01   8.642 5.59e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 670.4 on 25 degrees of freedom
Multiple R-squared: 0.9621, Adjusted R-squared: 0.9576
F-statistic: 211.5 on 3 and 25 DF,  p-value: < 2.2e-16
```





Feature	Value
RM	7.27
LSTAT	6.05
TAX	254.00
INDUS	6.41
PTRATIO	17.60

Model Specific vs. Model Agnostic

- **Model Specific:**

Transparency method's utility is limited to a specific model family

- Interpretable models (e.g. Weights in linear models)

- Using specific model internals (e.g. gradient)

Model Agnostic:

The method is can be used across many different model families.

- Usually works as: Input → **method** → output

- e.g. SHAP, LIME etc.

Local vs. Global Explanations

- **Local:**

Methods and techniques that explain individual predictions

- Important for case-by-case analysis

- credit reports, assessment feedback, law enforcement

Global:

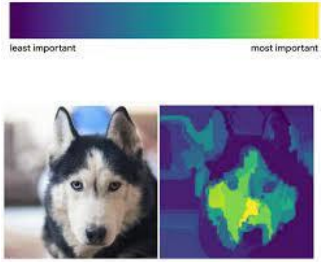
Methods that explain the behaviour of the entire model

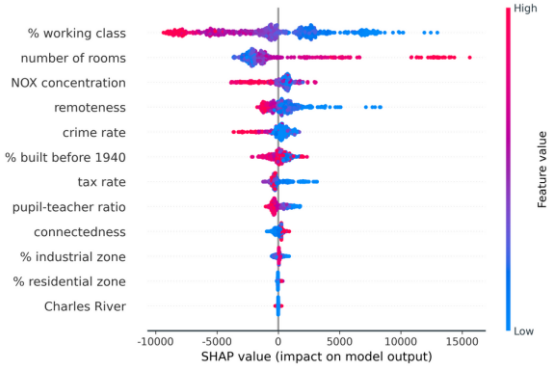
- Importance towards understanding the general behaviour of the model

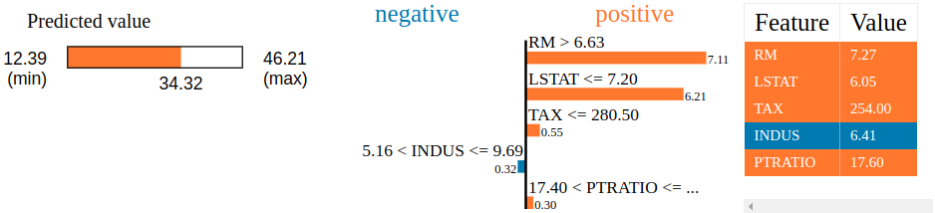
- e.g. what factors impact house price prediction?

Explainable AI Approaches

- Popular approaches revolve around features summaries and visualisations
- **What features does the model rely on?**
- Several Methods
  - Permutation Feature Importance
  - SHAP
  - LIME







Permutation Feature Importance

- Hypothesis: Changes of values in the more influential features lead to large fluctuations in the model outcomes.
- The model is sensitive to value changes in the influential features.

Permutation Feature Importance

	Feature_1	Feature_2	Feature_3	Label
Sample_1	1970	10.5	1	403.12
Sample_2	2020	14.9	2	412.15
Sample_3	1910	17.7	3	564.46

Permute "*Feature_2*"

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Sample_1	1970	17.7	1	403.12
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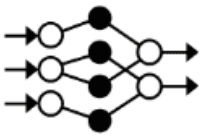
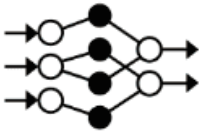


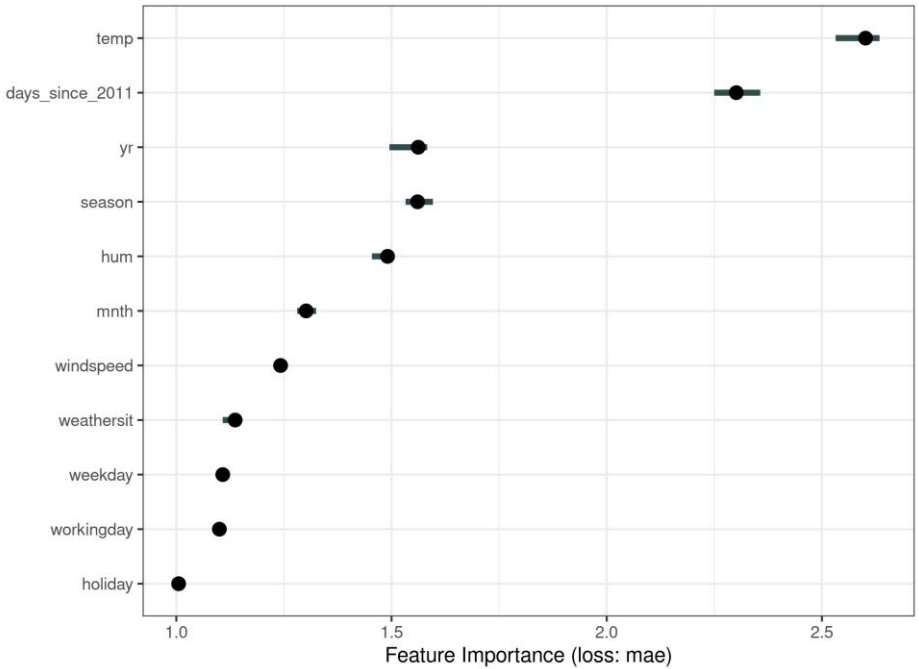


Price (Predicted Original)
350.54
420.24
600.13

Price (Predicted Permuted)
700.65
300.13
121.15

Average difference
= Feature Importance

Permutation Feature Importance



SHAP

Shapely Vales

- Come from Game Theory in Economics
- Objective: In Team efforts, we need to assign credit to the team members fairly.
- Different team members bring in different skills
- Rather than assuming all team players always Contribute, you also account for dynamics when some team members are missing.

  - E.g. All avengers don’t turn up to every fight.

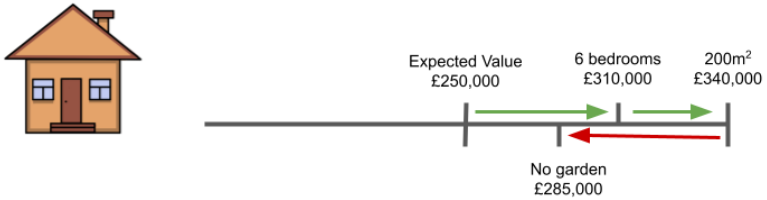
- So, what to do about the quarterly performance review?

Shapely Vales Example

- Let N denote the set of players, n the number of players, and S denote a particular combination of the players (a coalition)
  - $N = \{IM, CA, BW, H\}$ and $n = 4$
  - One possible coalition: $S = \{IM, BW\}$
- Suppose we can simulate the winnings $y(S)$ for all different coalitions S:
  - Examples: $y(\{IM, CA, BW, H\}) = 17$, $y(\{IM, CA\})$ = 9 and $y(BW) = 2$
- We can compute the difference in winnings, the **marginal contributions**, when Iron Man is excluded from the team:
  - $(y(S \cup \{IM\}) - y(S))$ for coalitions S that *exclude* Iron Man $S \subseteq N \backslash \{IM\}$
- Iron Man's Shapley Value is a **weighted average** of **all the marginal contributions**:
  - $\varphi_{IM} = \sum_{S \subseteq N \backslash \{IM\}} \frac{|S|!(n-|S|-1)!}{n!}(y(S \cup \{IM\}) - y(S))$
- Repeat the process for each avenger to get their Shapley Values

Shapely Vales for Features

- Features are the avengers
- Suppose we have a model f that predicts house price y and independent variables x1, x2 and x3, number of bedrooms, floor area and if the property has a garden.
- Assume average house price, E[y] = 250,000
- If we are to put a new house in the market, best uninformed guess is 250,000
- Our prediction updates when we know more about the house.



Computing Shapely Vales for Features

- Deal with "missing" features in the model by using the mean value E[x]
- Compute the model predictions for the different stages (combinations):
  - f(x1 = E[x1], x2 = E[x2], x3 = E[x3]) = 250,000
  - f(x1 = 6, x2 = E[x2], x3 = E[x3]) = 310,000
  - f(x1 = 6, x2 = 200, x3 = E[x3]) = 340,000
  - f(x1 = 6, x2 = 200, x3 = 0) = 285,000

Computing Marginal Contribution

- Marginal Contribution of x1 to the model is:
- Compute the effect of presence of x1 in observation x1=6, x2=200 and x3=0:

  - f(x1 = **6**, x2 = E[x2], x3 = E[x3]) - f(x1 = **E[x1],** x2 = E[x2], x3 = E[x3])
  - f(x1 = **6**, x2 = 200, x3 = E[x3]) = f(x1 = **E[x1]**, x2 = 200, x3 = E[x3])
  - f(x1 = **6**, x2 = E[x2], x3 =0) = f(x1 = **E[x1]**, x2 = E[x2], x3 =0)
  - f(x1 = **6**, x2 = 200, x3 = 0)= f(x1 = **E[x1]**, x2 = 200, x3 = 0)

SHAP Algorithm

- Inputs: trained model $f(x)$ and test data $X \in \mathbb{R}^{n \times p}$.
- Notation: let S be the set of feature values that are included in the model (rather than substituting for the expected value) and P is the set of all features

1. For each sample $i = 1, 2, \ldots, n$:
   A. For each feature $j = 1, 2, \ldots, p$:
      a. Compute the weighted average of the marginal contributions to get the Shapley Value φ_{ij} for sample i and feature j:

$$\varphi_{ij} = \sum_{S \subseteq P \setminus \{x_{ij}\}} \frac{|S|!(p - |S| - 1)!}{p!} (f(S \cup \{x_{ij}\}) - f(S))$$

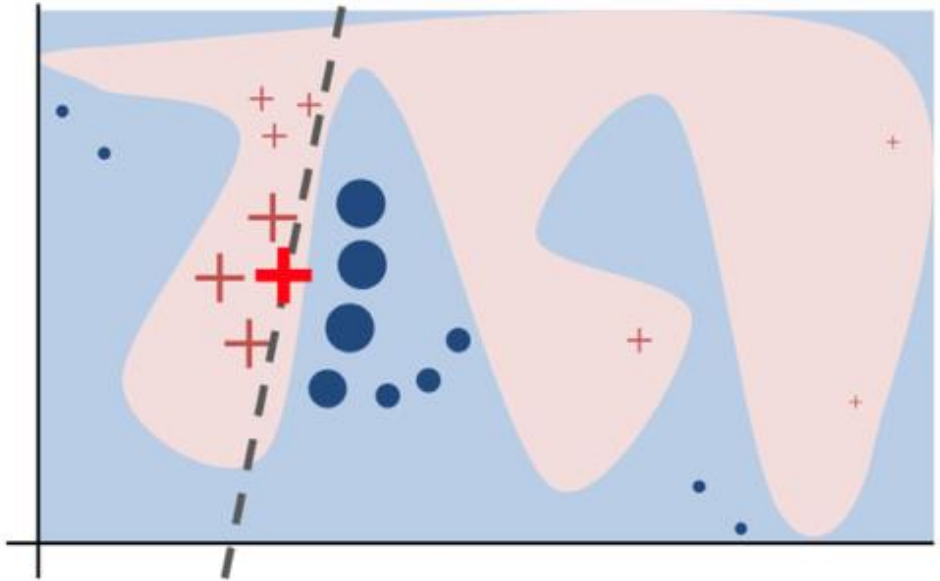
   B. The Shapley Values $\{\varphi_{i1}, \varphi_{i2}, \ldots, \varphi_{ip}\}$ give the **local SHAP explanation** for sample i
2. To compute the **global SHAP explanation**, for each feature $j = 1, 2, \ldots, p$:
   A. Compute the average of the magnitude of the local Shapley Values $\varphi_j = \frac{1}{n} \sum_{i=1}^{n} |\varphi_{ij}|$

LIME

- **L**ocal **I**nterpretable **M**odel-Agnostic **E**xplanations
- Idea
  - Given a black box model f and an observation x
  - Use a simpler model g to see how f behaves around x
  - Around x is defined by perturbed data points x’, x’’, x’’’, …
  - g comes from a humanly intuitive model family (e.g. linear model)

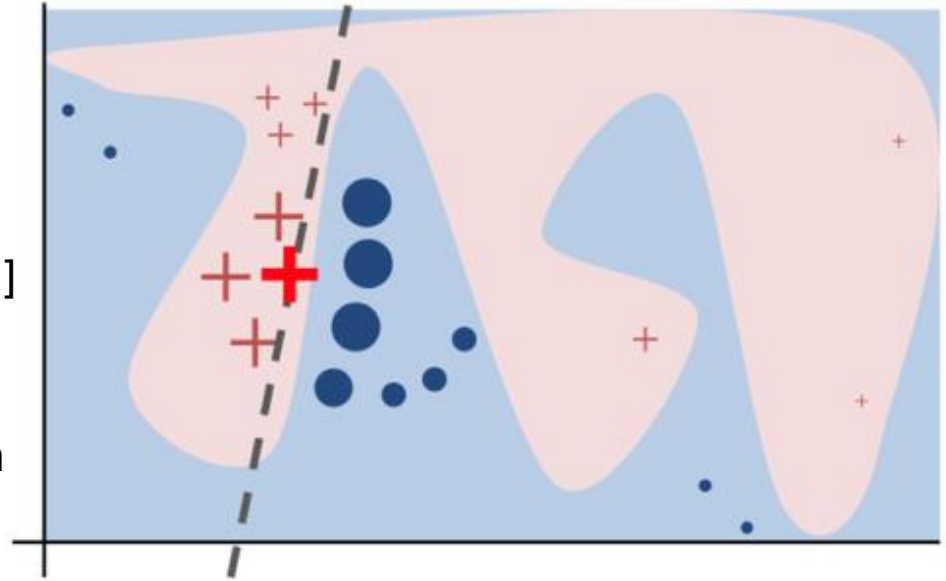
LIME

- **Key Assumption: One can find a linear approximator to a local area of a very complex mapping landscape when you zoom in enough**



LIME

- Select observation x
- Sample data points around x’ using data perturbation [x’, x’’, x’’’, …]
- Use f to get predictions for the perturbed points [(x’, f(x’), (x’’, f(x’’)),…]
- Weight new points based on its distance to x
- Use new weighted data points to train g
- Use g to explain behaviour of f near x



Weighting points

- One could use a kernel function
- Eg. Exponential Kernel

$$\pi_x(x') = \exp(\frac{-D(x, x')^2}{\sigma^2})$$

- We find a simpler g

$$g(x') = w_g^T . x'$$

- Loss function to find optimal g:

$$L(f, g; \pi_x) = \sum_i \pi_x(x_i')(f(x_i') - g(x_i'))^2$$

Summary

- The current AI landscape is filled with Black Box models
- We need explainability to inform stakeholders about the behaviour of black box models
- Feature importance is one of the most popular explainability techniques
- Feature permutation importance allow identify most influential features used by a model
- SHAP analysis allows identifying the importance of a feature in predicting an individual data point using game theory
- LIME an alternative approach builds a simpler, surrogate model for a local area in the data space to derive explanations.

Thank You

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